LEARNING PLAN – MANAGE DATABASE SYSTEMS

1. INSTITUTIONAL DETAILS

ITEM	DETAILS
Unit of Competence	Manage Computerized Database Systems
Unit Code	0612 451 08A
Name of Trainer	JAMES NDIRANGU
Competence Level	LEVEL 5
Institution	THE ABERDARE TECHNICAL COLLEGE
Number of Learners	3
Date of Preparation	06/08/2026
Skill or Job Task	Designing computerized databases, creating computerized databases, manipulating computerized databases, testing computerized databases and maintaining computerized databases.

2. BENCHMARK / CRITERIA TO BE USED

1. Design computerized database

1.1 Computerized database requirements are established as per user requirements and workplace procedures.
1.2 Appropriate database models are selected according to database requirements.
1.3 Conceptual database design is developed according to user requirements.
1.4 Logical database design is developed according to the conceptual design.
1.5 Physical database design is developed according to the logical design and user requirements.

2. Create computerized database

2.1 Database objects are created according to the database design.
2.2 Tables and fields are created according to database requirements.
2.3 Tables are linked according to identified relationships.
2.4 Queries, forms and reports are created according to user requirements.
2.5 Integrity constraints are implemented according to database requirements.
2.6 Database indexing is implemented according to database requirements.

3. Manipulate computerized database

3.1 Database is populated according to database requirements.
3.2 Database operations are performed according to user requirements.
3.3 Database queries are developed using appropriate SQL commands.
3.4 Database views are created and managed according to requirements.
3.5 Database reports are generated, edited, formatted and printed according to user requirements.

4. Test computerized database

4.1 Database test plan is developed according to database requirements.
4.2 Appropriate database test data is prepared according to the test plan.
4.3 Database testing is performed using appropriate test data.
4.4 Database results are validated against expected results.
4.5 Database test report is prepared according to workplace requirements.

5. Maintain computerized database

5.1 Database optimization techniques are applied according to performance requirements.
5.2 Computerized database maintenance schedule is prepared according to workplace procedures.
5.3 Computerized database is monitored according to the maintenance schedule.
5.4 Database backup and recovery procedures are performed according to workplace requirements.
5.5 Computerized database maintenance report is prepared according to workplace procedures.

LEARNING OUTCOME 1: DESIGN COMPUTERIZED DATABASE

WEEK 1

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
1	**1. Design compute**	**Session 1: Introduction to DBMS and**	**By the end of the session, trainees**	1.**Introduces** DBMS concepts and **poses**	1. **Listen** to the DBMS introduction and **answer**	Computers,	Oral questions,	Date: ________Reflection: ________

	rized database	**Database Requirements**• Introduction to DBMS• Definition of terms• Functions of database systems• Database applications • Types of DBMS• Major components of DBMS environment	**should be able to:** 1. **Define** database and DBMS terminology .2. **Explain** the functions of database systems.3. **Identify** database applications .4. **Classify** different types of DBMS.5. **Identify** the major components of a DBMS environment.	questions on basic database terminology. 2.**Explains** database terms and **demonstrates** their application. 3.**Demonstrates** common database applications and **asks** trainees to identify them.4. **Explains** different types of DBMS and **asks** trainees to classify examples.	questions on database terminology .2. **Define** database terms and **give** relevant examples. 3. **Identify** database applications demonstrated by the trainer. 4. **Classify** DBMS examples according to their types.5. **Identify** the major DBMS components demonstrated by the trainer.	projector, DBMS software, Notes	terminology exercise,	

				5.**Demonstrates** major DBMS components and **asks** trainees to identify them.		Curriculum Occupational standards	class discussion	
1	**1. Design computerized database**	**Session 2:** **Database Architecture and Requirements Gathering**• Database architecture • Requirements gathering techniques• Requirement elicitation• Database assessment requirement report	**By the end of the session, trainees should be able to**: 1. **Describe** database architecture .2. **Identify** requirements gathering techniques. 3. **Apply** requirement elicitation techniques. 4. **Analyse** user	1. **Explains** database architectures and **poses** questions on their differences .2.**Introduces** interviews, questionnaires and observation and **asks** trainees to identify	1. **Describe** database architectures and **answer** questions on their differences. 2. **Identify** interviews, questionnaires and observation techniques. 3. **Apply** requirement elicitation techniques during the	Computers projector, sample questionnaires,	Requirement elicitation exercise, requirement report	Date: ________Reflection: ________

			requirements. 5. **Prepare** a database assessment requirement report.	each technique. 3.**Demonstrates** requirement elicitation and **guides** trainees through a sample exercise .4.**Provides** user requirements and **guides** trainees in analysing them. 5.**Demonstrates** report preparation and **guides** trainees in preparing a requirement report.	sample exercise. 4. **Analyse** the provided user requirements. 5. **Prepare** a database assessment requirement report.	interview forms, report templates		

WEEK 2

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
2	**1. Design computerized database**	**Session 3: Database Models•** Relational model• Hierarchical model• Network model• Object-oriented model• Dimensional model	By the end of the session, trainees should be able to:1. **Identify** different database models.2. **Explain** the characteristics of each model.3. **Differentiate** database models.4. **Analyse** suitable models for	1.**Introduces** database models and **poses** identification questions. 2.**Explains** characteristics of each model and **asks** trainees to describe them.3. **Compares** database models and **asks** trainees to differentiate them.4.	1. **Identify** the database models and **answer** identification questions.2. **Describe** the characteristics of each model.3. **Differentiate** the database models using their characteristics.4. **Analyse** the application scenarios provided.5. **Select**	Database diagrams, computers, projector, DBMS software	Model comparison exercise, oral questions, case study	Date: ________Reflection: ________

			given applications.5. **Select** an appropriate database model for a given requirement.	**Provides** application scenarios and **guides** trainees in analysing them. 5. **Presents** database requirements and **guides** trainees in selecting suitable models.	suitable database models for the given requirements.			
2	**1. Design computerized database**	**Session 4: Conceptual Database Design and ERDs•** Conceptual design• ERDs• Types of	**By the end of the session, trainees should be able to**:1. **Explain** conceptual database design.2. **Identify** entities and	1. **Explains** conceptual database design and **poses** questions on its purpose.2. **Demonstrates** entities and	1. **Explain** conceptual database design and **answer** questions on its purpose.2. **Identify** entities and attributes from the	Computers, draw.io, ERD templates, projector	ERD drawing practical	Date: ________Reflection: ________

		entities and notations• Types of attributes and notations• Types of relationships and notations	attributes.3. **Differentiate** relationship types.4. **Construct** an ERD.5. **Evaluate** an ERD against user requirements.	attributes and **asks** trainees to identify them.3. **Explains** relationship types and **asks** trainees to differentiate them.4. **Demonstrates** ERD construction and **guides** trainees in drawing ERDs.5. **Provides** sample ERDs and **guides** trainees in evaluating them.	examples provided.3. **Differentiate** relationship types using the given examples.4. **Construct** ERDs using the demonstrated notation.5. **Evaluate** sample ERDs against the stated requirements.			

WEEK 3

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
3	**1. Design computerized database**	**Session 5: Relational and Logical Database Design**• Relational model• Logical design• Defining tables• Keys• Normalization	**By the end of the session, trainees should be able to**: 1.**Explain** relational database design.2. **Identify** tables and keys.3. **Apply** normalization principles. 4. **Analyse** unnormalized data.5.	1. **Explains** relational database design and **poses** questions on its principles. 2. **Demonstrates** tables and keys and **asks** trainees to identify them.3. **Demonstrates** normalization and **guides** trainees	1. **Explain** relational database design and **answer** questions on its principles.2. **Identify** tables and keys from the examples provided.3. **Apply** normalization principles to the given data.4. **Analyse** the unnormalized data and	Computers, DBMS, normalization worksheets, ERD diagrams	Normalization exercise, logical design practical	Date: ________Reflection: ________

			Develop a logical database design.	through normalization exercises.4. **Provides** unnormalized data and **guides** trainees in analysing it.5. **Demonstrates** conversion of an ERD into tables and **guides** logical design development.	identify problems.5. **Develop** a logical database design from the ERD.			
3	**1. Design computerized database**	**Session 6: Physical Database Design•** Selection of data types•	By the end of the session, trainees should be able to:1. **Identify**	1. **Introduces** database data types and **poses** selection questions.	1. **Identify** appropriate data types and **answer** selection questions.2. **Select**	Computers, DBMS software, database design worksheets	Physical design practical	Date: ________Reflection: ________

		Indexing• Constraints	appropriate data types.2. **Select** suitable data types for fields.3. **Explain** database indexing.4. **Apply** database constraints.5. **Develop** a physical database design.	2. **Demonstrates** data type selection and **asks** trainees to select appropriate types.3. **Explains** database indexing and **demonstrates** index creation.4. **Demonstrates** database constraints and **guides** trainees in applying them.5. **Provides** a database design task and **guides**	suitable data types for the given fields.3. **Explain** indexing and **create** indexes as demonstrated.4. **Apply** the appropriate database constraints. 5. **Develop** a physical database design from the logical design.			

				trainees in developing the physical design.				

WEEK 4

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
4	**1. Design computerized database**	**Session 7: Integrated Database Design•** Requirements• Database model• Conceptual design• ERD• Logical design• Physical design	**By the end of the session, trainees should be able to**:1. **Analyse** database requirements.2. **Evaluate** alternative database models.3. **Develop** an ERD.4.	1. **Provides** a database case study and **guides** trainees in analysing requirements.2. **Presents** alternative database models and **asks** trainees to evaluate them.3.	1. **Analyse** the database case study and **identify** the requirements.2. **Evaluate** the alternative database models and **select** the appropriate model.3.	Computers, draw.io, DBMS, case studies	**CAT 1: Complete computerized database design practical**	Date: ________Reflection: ________

			Create logical tables and keys.5. **Create** a physical database design.6. **Evaluate** the complete design against user requirements.	**Demonstrates** ERD development and **guides** trainees in developing ERDs.4. **Demonstrates** conversion of ERDs into logical tables and **guides** trainees in creating tables and keys.5. **Demonstrates** physical design development and **guides** trainees in creating the design.6. **Reviews**	**Develop** an ERD from the identified requirements.4. **Create** logical tables and keys from the ERD.5. **Create** the physical database design.6. **Evaluate** the completed design against the user requirements.			

				completed designs and **asks** trainees to evaluate them against requirements.				

LEARNING OUTCOME 2: CREATE COMPUTERIZED DATABASE

WEEK 5

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
5	**2. Create computerized database**	**Session 8: Database Objects, Fields and Data Types•** Tables• Forms• Queries•	**By the end of the session, trainees should be able to**:1. **Identify** database objects.2.	1. **Introduces** database objects and **asks** trainees to identify them.2. **Explains**	1. **Identify** database objects and **answer** identification questions.2. **Explain** the purpose	Computers, MS Access/DB MS software, projector	Database object identification, practical exercise	Date: ________Reflection: ________

		Reports• Macros• Fields• Records• Data types• Field properties • Keys	**Explain** the purpose of each database object.3. **Differentiate** fields and records.4. **Select** appropriate data types.5. **Apply** field properties and keys.	the purpose of each object and **poses** questions on their uses.3. **Demonstrates** fields and records and **asks** trainees to differentiate them.4. **Demonstrates** data type selection and **guides** trainees in selecting types.5. **Demonstrates** field properties and keys and **guides** trainees in	of each database object.3. **Differentiate** fields from records using the examples provided.4. **Select** appropriate data types for given fields.5. **Apply** field properties and keys as demonstrated.			

				applying them.				
5	**2. Create computerized database**	**Session 9: Creation and Linking of Tables•** Creation of tables• Primary keys• Foreign keys• Linking tables• Relationships	By the end of the session, trainees should be able to:1. **Create** database tables.2. **Define** primary keys.3. **Create** foreign keys.4. **Link** tables.5. **Evaluate** table relationships.	1. **Demonstrates** table creation and **guides** trainees in creating tables.2. **Demonstrates** primary key creation and **asks** trainees to define primary keys.3. **Demonstrates** foreign key creation and **guides** trainees in creating	1. **Create** tables following the demonstrated procedure. 2. **Define** primary keys for the created tables.3. **Create** appropriate foreign keys.4. **Link** tables using the identified relationships.5. **Evaluate** the table relationships and	Computers, DBMS, sample datasets	Table creation and linking practical	Date: ________Reflection: ________

				foreign keys.4. **Demonstrates** table linking and **guides** trainees in linking tables.5. **Provides** relationship examples and **asks** trainees to evaluate them.	correct errors.			

WEEK 6

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
6	**2. Create computerized**	**Session 10: Database Development –**	By the end of the session, trainees should be	1. **Explains** queries, forms and reports and	1. **Explain** the purposes of queries, forms and reports and	Computers, DBMS, sample database	Queries, forms and reports practical	Date: ________Reflection: ________

	database	**Queries, Forms and Reports**• Creation of queries• Creation of forms• Creation of reports	able to:1. **Explain** the purpose of queries, forms and reports.2. **Create** database queries.3. **Create** database forms.4. **Create** database reports.5. **Evaluate** database objects against user requirements.	**poses** questions on their purposes.2. **Demonstrates** query creation and **guides** trainees in creating queries.3. **Demonstrates** form creation and **guides** trainees in designing forms.4. **Demonstrates** report creation and **guides** trainees in generating reports.5. **Provides** user requirements and **asks**	**answer** questions.2. **Create** queries according to the demonstrated procedure.3. **Create** forms according to the given requirements.4. **Create** reports using the sample database.5. **Evaluate** the completed objects against user requirements.			

				trainees to evaluate their objects.				
6	**2. Create computerized database**	**Session 11: Integrity Constraints and Database Indexing•** Nulls• Entity integrity• Referential integrity• General constraints • Primary indexing• Secondary indexing	By the end of the session, trainees should be able to:1. **Explain** database integrity.2. **Identify** types of integrity constraints.3. **Apply** entity and referential integrity.4. **Create** primary and secondary indexes.5. **Evaluate**	1. **Explains** database integrity and **poses** questions on its importance.2. **Demonstrates** integrity constraints and **asks** trainees to identify them.3. **Demonstrates** entity and referential integrity and **guides** trainees in	1. **Explain** database integrity and **answer** questions on its importance. 2. **Identify** the different integrity constraints demonstrated.3. **Apply** entity and referential integrity to the database.4. **Create** primary and secondary indexes.5. **Evaluate** database	Computers, DBMS, sample database	Integrity and indexing practical	Date: ________Reflection: ________

			database integrity.	applying them.4. **Demonstrates** primary and secondary indexing and **guides** trainees in creating indexes.5. **Provides** integrity scenarios and **asks** trainees to evaluate them.	integrity and identify violations.			

WEEK 7

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
7	**2. Create computerized database**	**Session 12: Data Relationships**• One-to-one• One-to-many• Many-to-one• Many-to-many relationships	By the end of the session, trainees should be able to:1. **Identify** database relationships.2. **Explain** relationship types.3. **Differentiate** one-to-one, one-to-many, many-to-one and many-to-many relationshi	1. **Introduces** database relationships and **asks** trainees to identify examples.2. **Explains** relationship types and **poses** questions about their characteristics.3. **Compares** relationship types and **asks** trainees to differentiate	1. **Identify** database relationships and **give** relevant examples.2. **Explain** the characteristics of each relationship type.3. **Differentiate** the relationship types using the given scenarios.4. **Apply** relationships when	Computers, DBMS, ERDs, sample databases	Relationship practical	Date: ________Reflection: ________

			ps.4. **Apply** relationships when linking tables.5. **Evaluate** relationships for correctness.	them.4. **Demonstrates** relationship implementation and **guides** trainees in linking tables.5. **Provides** relationship scenarios and **asks** trainees to evaluate them.	linking database tables.5. **Evaluate** the relationships and correct errors.			
7	**2. Create computerized database**	**Session 13: Integrated Database Creation•** Tables• Relationships• Queries• Forms• Reports•	By the end of the session, trainees should be able to:1. **Create** database tables.2. **Link** database	1. **Demonstrates** complete database creation and **guides** trainees in creating tables.2. **Demonstrates**	1. **Create** database tables from the provided design.2. **Link** tables using appropriate relationships.3.	Computers, DBMS, database design, sample data	**CAT 2 + Practical Assessment 1: Database Creation**	Date: ________Reflection: ________

		Constraints • Indexing	tables.3. **Create** queries, forms and reports.4. **Apply** integrity constraints. 5. **Implement** database indexing.6. **Evaluate** the completed database.	relationships and **guides** trainees in linking tables.3. **Demonstrates** queries, forms and reports and **guides** trainees in creating them.4. **Demonstrates** integrity constraints and **guides** trainees in applying them.5. **Demonstrates** indexing and **guides** trainees in implementing indexes.6. **Reviews** completed databases	**Create** queries, forms and reports.4. **Apply** appropriate integrity constraints. 5. **Implement** suitable database indexes.6. **Evaluate** the completed database against the requirements.			

				and **asks** trainees to evaluate them.				

LEARNING OUTCOME 3: MANIPULATE COMPUTERIZED DATABASE

WEEK 8

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
8	**3. Manipulate computerized database**	**Session 14: Database Population and Operations**• Database population• INSERT• SELECT • UPDATE	By the end of the session, trainees should be able to:1. **Explain** database population.2. **Insert** data into a database. 3. **Select**	1. **Explains** database population and **poses** questions on its purpose.2. **Demonstrates** INSERT operations and **guides** trainees in	1. **Explain** database population and **answer** questions on its purpose.2. **Insert** data into the database as demonstrated.3. **Select** required records	Computers, DBMS, SQL environment, sample database	Database manipulation practical	Date: ________Reflection: ________

		• DELETE	required records.4. **Update** existing records.5. **Delete** records correctly. 6. **Evaluate** database operations .	inserting data.3. **Demonstrates** SELECT operations and **guides** trainees in retrieving records.4. **Demonstrates** UPDATE operations and **guides** trainees in modifying records.5. **Demonstrates** DELETE operations and **guides** trainees in deleting records safely.6. **Provides** database	from the database.4. **Update** existing database records.5. **Delete** specified records correctly.6. **Evaluate** the results of the database operations.			

				operations and **asks** trainees to evaluate the results.				
8	**3. Manipulate computerized database**	**Session 15: SQL and Database Querying•** Introduction to SQL• Writing SQL commands• SQL views	By the end of the session, trainees should be able to:1. **Define** SQL.2. **Explain** common SQL commands.3. **Write** basic SQL commands.4. **Apply** SQL to retrieve and manipulate data.5.	1. **Introduces** SQL and **poses** questions about its purpose.2. **Explains** common SQL commands and **asks** trainees to identify their uses.3. **Demonstrates** SQL syntax and **guides** trainees in writing commands.	1. **Define** SQL and **answer** questions about its purpose.2. **Identify** common SQL commands and **state** their uses.3. **Write** SQL commands using the demonstrated syntax.4. **Apply** SQL commands to retrieve and manipulate data.5.	Computers, MySQL/SQL Server/Access, SQL worksheets	SQL practical exercise	Date: ________Reflection: ________

			Create SQL views.	4. **Demonstrates** data retrieval and manipulation and **guides** trainees in executing SQL.5. **Demonstrates** SQL view creation and **guides** trainees in creating views.	**Create** SQL views according to the given requirements.			

WEEK 9

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
9	**3. Manipulate computerized database**	**Session 16: Database Report Generation** • Generate report• Edit report• Format report• Apply header and footer• Print report	By the end of the session, trainees should be able to:1. **Explain** database report generation.2. **Generate** database reports.3. **Edit** database reports.4. **Format** database reports.5. **Apply** headers	1. **Explains** report generation and **poses** questions on its purpose.2. **Demonstrates** report generation and **guides** trainees in generating reports.3. **Demonstrates** report editing and **guides** trainees in editing reports.4.	1. **Explain** report generation and **answer** questions on its purpose.2. **Generate** database reports from the sample database.3. **Edit** the generated reports.4. **Format** reports according to the requirements.5. **Apply**	Computers, DBMS, printer, report templates	Report generation practical	Date: ________Reflection: ________

			and footers.6. **Produce** professional printed reports.	**Demonstrates** report formatting and **guides** trainees in formatting reports.5. **Demonstrates** headers and footers and **guides** trainees in applying them.6. **Demonstrates** report printing and **guides** trainees in producing final reports.	appropriate headers and footers.6. **Produce** and print professional reports.			
9	**3. Manipulate computerized**	**Session 17: Integrated Database Manipulation**• Data	By the end of the session, trainees should be	1. **Provides** a practical scenario and **guides**	1. **Apply** database operations to the practical	Computers, DBMS, SQL, sample	**CAT 3: Database Manipulation Practical**	Date: ________Reflection: ________

	databa se	population• Database operations• SQL• Views• Reports	able to:1. **Apply** database operation s.2. **Write** SQL queries.3. **Create** database views.4. **Generate and format** database reports.5. **Analyse** query results.6. **Evaluate** database manipulat ion activities.	trainees in applying database operations. 2. **Provides** SQL tasks and **guides** trainees in writing queries.3. **Demonstr ates** view creation and **guides** trainees in creating views.4. **Demonstr ates** report generation and **guides** trainees in generating and formatting reports.5. **Provides** query	scenario.2. **Write** and execute SQL queries.3. **Create** database views from the given requirement s.4. **Generate and format** database reports.5. **Analyse** the query results and identify findings.6. **Evaluate** the completed database manipulatio n activities.	database , printer		

				results and **asks** trainees to analyse them.6. **Reviews** the completed activities and **asks** trainees to evaluate their work.				

LEARNING OUTCOME 4: TEST COMPUTERIZED DATABASE

WEEK 10

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
10	**4. Test computerized database**	**Session 18: Database Test Planning and Test**	By the end of the session, trainees should be	1. **Explains** the importance of database	1. **Explain** the importance of database testing and	Computers, DBMS, test template	Test plan and test-data preparati	Date: ________Reflection: ________

		Data• Importance of database testing• Types of database testing• Structural DB testing• Schema• Tables and columns• Functional testing• White-box testing• Black-box testing• Stored procedure testing• Non-functional testing• Load testing• Stress testing• Test data preparation	able to:1. **Explain** the importance of database testing.2. **Identify** database testing types.3. **Differentiate** structural, functional and non-functional testing.4. **Analyse** database testing requirements.5. **Prepare** appropriate test data.6. **Develop** a database test plan.	testing and **poses** questions on its purpose.2. **Introduces** database testing types and **asks** trainees to identify them.3. **Compares** structural, functional and non-functional testing and **asks** trainees to differentiate them.4. **Provides** database requirements and **guides** trainees in	**answer** questions.2. **Identify** the different database testing types.3. **Differentiate** structural, functional and non-functional testing.4. **Analyse** the database testing requirements provided.5. **Prepare** appropriate test data.6. **Develop** a database test plan.	s, sample databases	on exercise	

				analysing testing requirements.5. **Demonstrates** test-data preparation and **guides** trainees in preparing test data.6. **Demonstrates** test-plan development and **guides** trainees in developing a plan.				
10	**4. Test computerized database**	**Session 19: Performing Database Testing and Validation•** Database testing using	By the end of the session, trainees should be able to:1. **Apply**	1. **Demonstrates** database testing procedures and **guides**	1. **Apply** database testing procedures using the provided test cases.2.	Computers, DBMS, testing tools, test data,	Database testing practical	Date: ________Reflection: ________

		test data• Validation of database results• Structural testing• Functional testing• White-box testing• Black-box testing• Stored procedure testing• Load testing• Stress testing	database testing procedures. 2. **Perform** structural testing.3. **Perform** functional testing.4. **Perform** non-functional testing.5. **Validate** database results.6. **Evaluate** database performance against expected results.	trainees in applying them.2. **Demonstrates** structural testing and **guides** trainees in testing database structures.3. **Demonstrates** functional testing and **guides** trainees in testing functions.4. **Demonstrates** non-functional testing and **guides** trainees in carrying	**Perform** structural testing on the database.3. **Perform** functional testing on database functions.4. **Perform** appropriate non-functional tests.5. **Validate** actual results against expected results.6. **Evaluate** database performance and identify defects.	test cases		

				out appropriate tests.5. **Provides** expected and actual results and **guides** trainees in validating them.6. **Provides** performance results and **asks** trainees to evaluate database performance.				
10	**4. Test computerized database**	**Session 20: Database Test Report•** Importance of test reports• Test report preparation•	By the end of the session, trainees should be able to:1. **Explain** the	1. **Explains** the importance of test reports and **poses** questions	1. **Explain** the importance of test reports and **answer** questions.2. **Organize**	Computers, report templates, test results, projector	**CAT 4 / Database Testing and Test Report**	Date: ________Reflection: ________

		Test results• Defects identified• Analysis• Recommendations	importance of test reports.2. **Organize** test results.3. **Analyse** database test findings.4. **Prepare** a database test report.5. **Present** test findings.6. **Recommend** appropriate corrective measures.	on their purpose.2. **Demonstrates** organization of test results and **guides** trainees in organizing them.3. **Provides** test findings and **guides** trainees in analysing them.4. **Demonstrates** test report preparation and **guides** trainees in preparing reports.5. **Demonstrates** professiona	the provided test results.3. **Analyse** the database test findings.4. **Prepare** a database test report.5. **Present** the test findings professionally.6. **Recommend** appropriate corrective measures.		**Practical**	

				l presentation of findings and **guides** trainees in presenting their findings.6. **Discusses** identified defects and **asks** trainees to recommend corrective measures.				

LEARNING OUTCOME 5: MAINTAIN COMPUTERIZED DATABASE

WEEK 11

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
11	**5. Maintain computerized database**	**Session 21: Database Optimization•** Introduction to database optimization• Query optimization techniques • Indexing strategies• Partitioning and storage management	By the end of the session, trainees should be able to:1. **Explain** database optimization.2. **Identify** factors affecting database performance.3. **Analyse** poorly performing queries.4.	1. **Explains** database optimization and **poses** questions on its importance.2. **Identifies** performance factors and **asks** trainees to identify them in sample databases. 3.	1. **Explain** database optimization and **answer** questions on its importance.2. **Identify** factors affecting database performance.3. **Analyse** poorly performing queries.4. **Apply**	Computers, DBMS, sample databases, SQL tools	Query optimization practical	Date: ________Reflection: ________

			Apply query optimization techniques. 5. **Apply** appropriate indexing strategies.6. **Evaluate** partitioning and storage management strategies.	**Demonstrates** poorly performing queries and **guides** trainees in analysing them.4. **Demonstrates** query optimization techniques and **guides** trainees in applying them.5. **Demonstrates** indexing strategies and **guides** trainees in applying them.6. **Explains** partitioning and	query optimization techniques to improve performance.5. **Apply** suitable indexing strategies.6. **Evaluate** partitioning and storage management strategies.			

				storage management and **asks** trainees to evaluate the strategies.				
11	**5. Maintain computerized database**	**Session 22: Database Maintenance Schedule and Backup & Recovery•** Types of database maintenance techniques • Maintenance schedule preparation • Key	By the end of the session, trainees should be able to:1. **Identify** database maintenance techniques. 2. **Explain** the importance of maintenance schedules. 3. **Prepare** a database	1. **Explains** preventive, corrective and adaptive maintenance and **asks** trainees to identify each type.2. **Explains** the importance of maintenance	1. **Identify** preventive, corrective and adaptive maintenance techniques. 2. **Explain** the importance of maintenance schedules and **answer** questions.3. **Prepare** a database	Computers, DBMS, backup storage, maintenance templates	Maintenance schedule and backup/recovery practical	Date: ________Reflection: ________

		elements in maintenance schedule• Database backup and recovery	maintenance schedule.4. **Identify** key maintenance activities.5. **Perform** database backup.6. **Apply** database recovery procedures.	schedules and **poses** questions on their purpose.3. **Demonstrates** maintenance schedule preparation and **guides** trainees in preparing schedules. 4. **Identifies** key maintenance activities and **asks** trainees to identify them from scenarios. 5. **Demonstr**	maintenance schedule.4. **Identify** key maintenance activities from the given scenarios.5. **Perform** database backup using the demonstrated procedure. 6. **Apply** database recovery procedures and verify the recovered database.			

				ates database backup and **guides** trainees in performing backups.6. **Demonstrates** recovery procedures and **guides** trainees in recovering a database.				

WEEK 12

Week	Main Learning Outcome	Session Title / Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources	Assessment	Reflections and Dates
12	**5. Maintain computerized database**	**Session 23: Computerized Database Monitoring** • Database monitoring tools• Resource utilization• Query performance • Transaction log monitoring• Connection monitoring• Alerting systems	By the end of the session, trainees should be able to:1. **Identify** database monitoring tools.2. **Explain** database monitoring requirements.3. **Monitor** resource utilization. 4. **Analyse** query performance.5.	1. **Introduces** database monitoring tools and **asks** trainees to identify them.2. **Explains** monitoring requirements and **poses** questions on their importance.3. **Demonstrates** resource monitoring	1. **Identify** database monitoring tools and **answer** identification questions. 2. **Explain** database monitoring requirements and **answer** questions on their importance.3. **Monitor** database resource utilization	Computers, DBMS monitoring tools, sample database, monitoring software	Database monitoring practical	Date: ________Reflection: ________

			Monitor transaction logs and connections.6. **Configure** appropriate alerting mechanisms.	and **guides** trainees in monitoring resource utilization. 4. **Demonstrates** query performance monitoring and **guides** trainees in analysing performance.5. **Demonstrates** transaction log and connection monitoring and **guides** trainees in monitoring them.6. **Demonstrates** alert configurati	using the available tools.4. **Analyse** query performance and identify performance problems.5. **Monitor** transaction logs and database connections.6. **Configure** appropriate database alerts.			

				on and **guides** trainees in configuring alerts.				
12	**5. Maintain computerized database**	**Session 24: Database Maintenance Report and Integrated Maintenance**• Maintenance findings• Optimization results• Monitoring results• Backup/recovery records• Maintenance report	By the end of the session, trainees should be able to:1. **Analyse** database maintenance findings.2. **Evaluate** database performance.3. **Apply** corrective maintenance measures.4. **Organize** maintenance records.5. **Prepare** a	1. **Provides** maintenance findings and **guides** trainees in analysing them.2. **Provides** performance results and **asks** trainees to evaluate database performance.3. **Demonstrates** corrective maintenance measures	1. **Analyse** the database maintenance findings provided.2. **Evaluate** database performance using the given results.3. **Apply** appropriate corrective maintenance measures.4. **Organize** database maintenance	Computers, DBMS, maintenance records, report templates, projector	**Final Integrated Practical + Database Maintenance Report**	Date: ________Reflection: ________

			database maintenance report.6. **Present** maintenance findings professionally.	and **guides** trainees in applying them.4. **Demonstrates** maintenance record organization and **guides** trainees in organizing records.5. **Demonstrates** maintenance report preparation and **guides** trainees in preparing reports.6. **Demonstrates** professional presentation	records.5. **Prepare** a database maintenance report.6. **Present** the maintenance findings professionally.			

				techniques and **guides** trainees in presenting findings.				